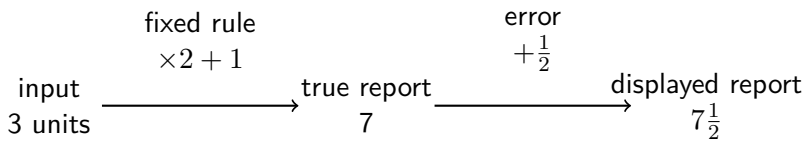
For Problems 6–7, the machine uses one fixed stretch and shift. The reference lengths are exact. Each displayed report may be up to 1 above or below its true report.



Problem 6: For each pair of reference strips, find the smallest and largest possible stretch. Give a machine that reaches each extreme.

Reference pair A	Reference pair B
2 units: display 8 4 units: display 12	2 units: display 8 10 units: display 24

Problem 7: Choose two exact reference lengths from 1 to 10 units before seeing any reports. Which pair keeps the range of possible stretches narrowest, however unhelpful the errors are? Could repeating one reference ten times guarantee as narrow a range?